

CSE490 (D) INTRODUCTION TO FOOTBALL DATA AND ANALYTICS

IN ASSOCIATION WITH BANGLADESH FOOTBALL FEDERATION

SPRING 2026



- Professional Certificate from BFF
- Internship Opportunity at **BFF Data Science** Department after successful completion
- Learning from the **industry experts** of Football Data Science
- Opportunity to **co-author** research papers for **Sports Data Conferences**

CSE490 (D) Introduction to Football Data & Analytics Lecture-8

Prepared by-
Tarvir Anjum Aditto
Lecturer,
CSE, Brac University

Evaluating Actions

- ❖ Recap position based xT calculation
- ❖ Self read: [Introducing Expected Threat \(xT\)](#) (included in syllabus)



Expected Threat (xT)- Recap

- Measure of the the affect of attacking actions
- **Position-based Expected Threat:**
where we assign a value to every position on the field
- **Action-based Expected Threat:**
where we assign a value to every action, based on its start and end position and various qualifiers describing that action

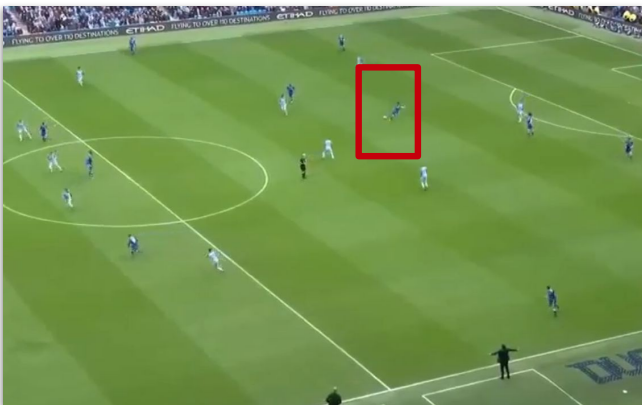


Expected Threat (xT)- Action Based

- An approach based on chains of possession.
- **Possession chain:** a sequence of events where the same team has unbroken control of the ball.
- Possession chain as the building block for an expected threat based on actions within that chain.
- **Why focussing on chain rather than position?** - The threat lies in how the ball is moved, rather than where the ball is.
- **Cross-field passes** can have positive value because they **shift the defence** and create space, even without advancing the ball.
- **Back passes** can also be valuable as they reset play and enable future forward progression.
- By looking at the **actions within a chain**, we can start to look at **context**.

Traditional statistics fall short as they fail to account for the context of the actions

Key Pass? **No**
Assist? **No**



Long ball from Fàbregas to Hazard

Key Pass? **Yes**
Assist? **Yes**



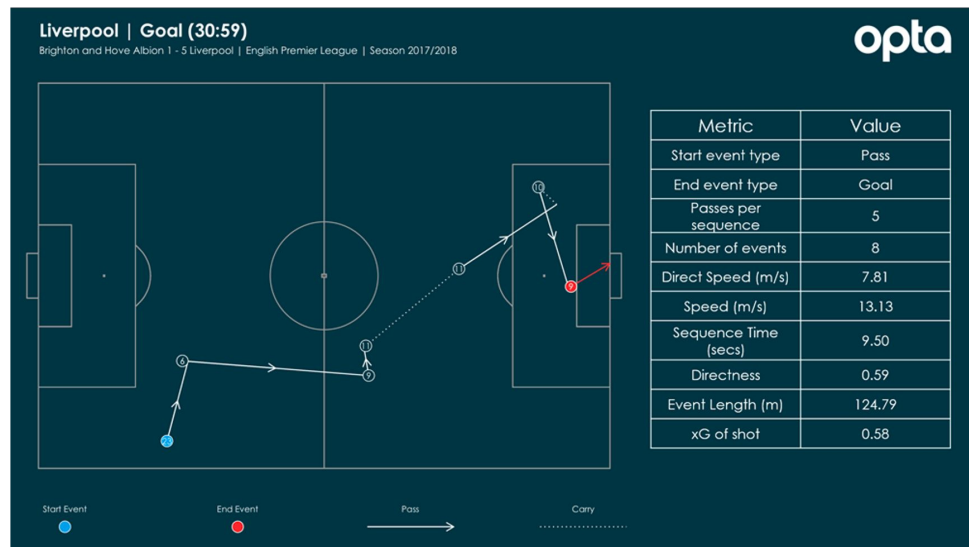
Short sideward pass from Busquets to Messi

Possession Chain

1. Find all the possessions each player is involved in.
2. Find all the shots within those possessions.
3. Sum their xG (you might take the highest xG per possession, or

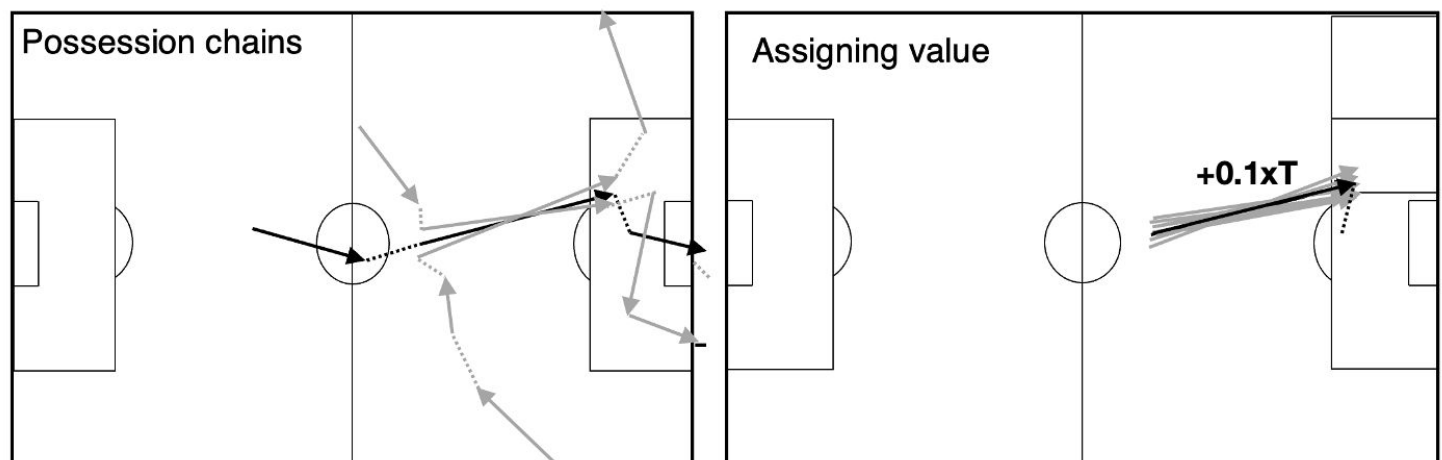
you might treat the shots as dependent events).

4. Assign that sum to each player, however involved they were.



Possession Chain

- All three chains have a similar pass.
- We might find, as in the example, below that one in ten of the chains containing a pass like this lead to a goal.



Evaluating Actions in a Chain

- ❖ Fit a model which predicts the expected goal value (output variable) of the chain as a function of the start and end co-ordinates of the pass (input variables).

$$P(\text{Shot} | x_1, y_1, x_2, y_2)$$

$$P(\text{Goal} | \text{Shot}, x_1, y_1, x_2, y_2)$$

Chain	x_1	y_1	x_2	y_2	Shot	xG
A	50	70	54	56	0	0
A	55	50	80	61	0	0
B	40	55	51	48	1	0.2
B	54	50	80	60	1	0.2
C	73	0	62	18	0	0
C	59	39	59	47	0	0
C	54	50	80	60	0	0

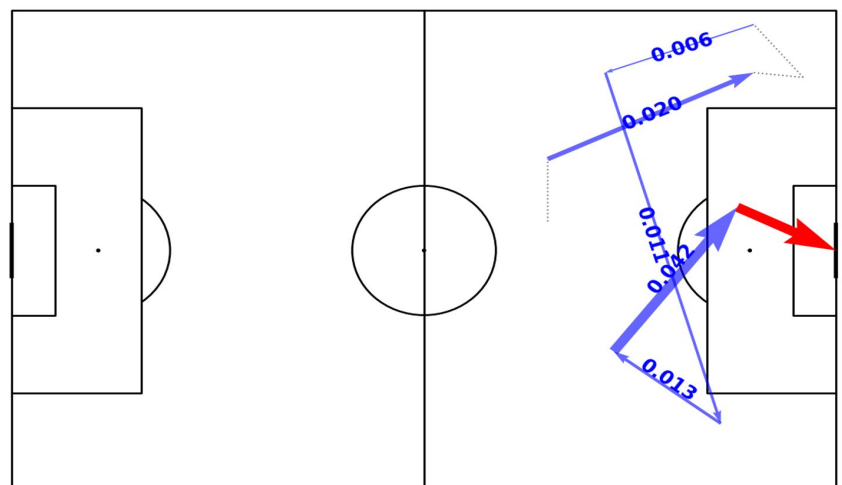
Evaluating Actions in a Chain

- ❖ Fit a model which predicts the expected goal value (output variable) of the chain as a function of the start and end co-ordinates of the pass (input variables).

$$P(\text{Shot} | x_1, y_1, x_2, y_2)$$

$$P(\text{Goal} | \text{Shot}, x_1, y_1, x_2, y_2)$$

Passes leading to a shot



VAEP

Valuing Actions by Estimating Probabilities

Very Important resource! : [Exploring VAEP](#)

Compute the difference between the values of the pre-action and post-action game states

Pre-action game state



Likelihood to **score** a goal = 0.03
Likelihood to **concede** a goal = 0.02



Post-action game state



Likelihood to **score** a goal = 0.05
Likelihood to **concede** a goal = 0.01

Score increase = 0.02
Concede decrease = 0.01
Contribution rating = **0.03**

Describe a game state as a sequence of 3 events

Game as sequence of ~1600 events

...	Dribble Neymar	Duel Neymar	Pass Neymar	Pass Busquets	Take-on Messi	Shot Messi	...
-----	-------------------	----------------	----------------	------------------	------------------	---------------	-----



Pre-action game state as subsequence of 3 events

	Dribble Neymar	Duel Neymar	Pass Neymar				
--	-------------------	----------------	----------------	--	--	--	--

Post-action game state as subsequence of 3 events

		Duel Neymar	Pass Neymar	Pass Busquets			
--	--	----------------	----------------	------------------	--	--	--

Describe a game state as a feature vector

	Dribble Neymar	Duel Neymar	Pass Neymar				
--	-------------------	----------------	----------------	--	--	--	--



- Start and end location
- Type of event
- Body part used
- Speed of play
- ...

Goal within next ten actions?



12	63	27	...	RF	58	16	27	...	RF	57	15	39	...	RF	87	-5	27	...	11	1
----	----	----	-----	----	----	----	----	-----	----	----	----	----	-----	----	----	----	----	-----	----	---

Dribble Neymar

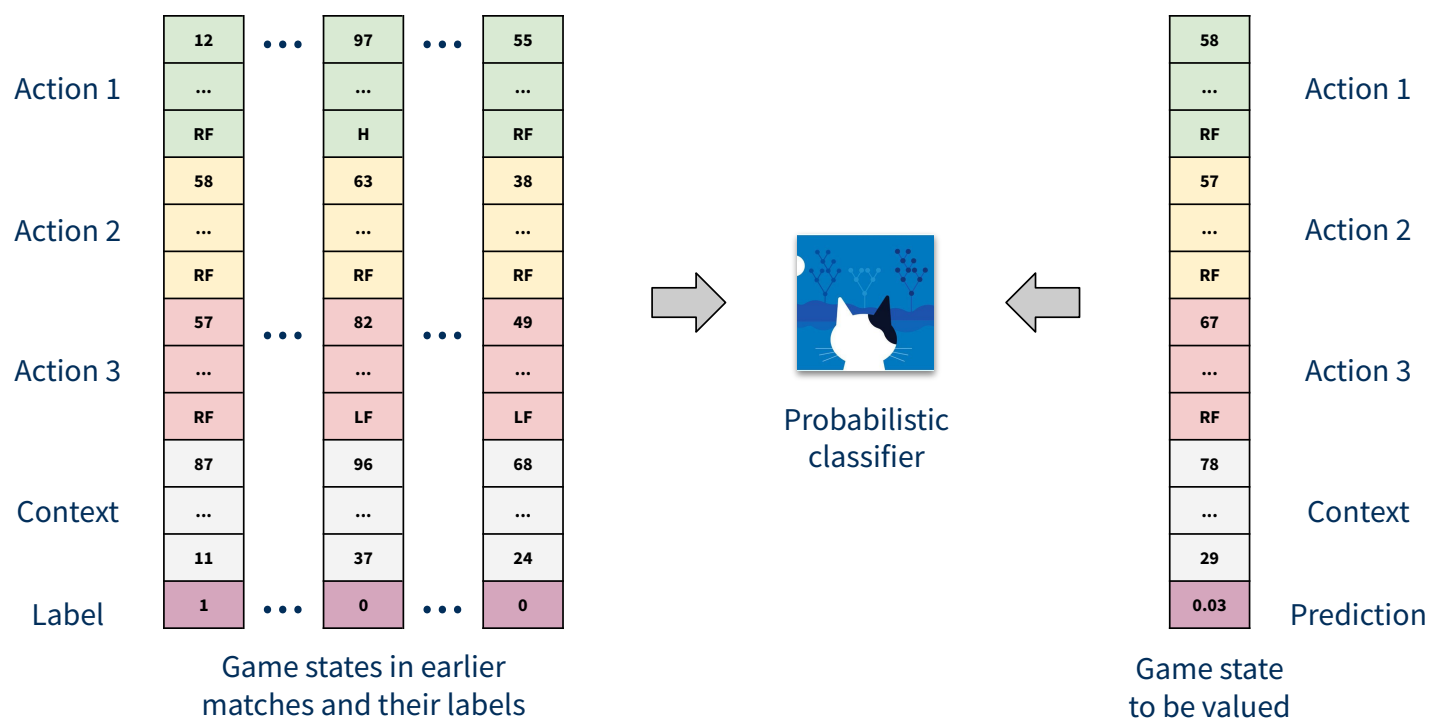
Duel Neymar

Pass Neymar

Contextual information

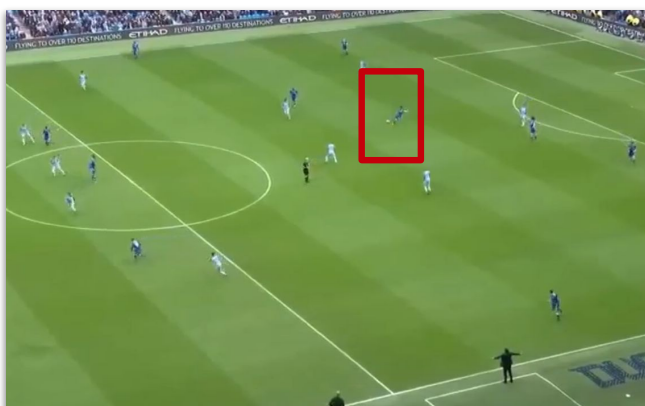
Label

Predict whether a game state will yield a goal



VAEP Ratings do account for the context of actions

"The Fàbregas pass is about 7 times more valuable than the Busquets pass"



Long ball from Fàbregas to Hazard



Short sideward pass from Busquets to Messi

Comparison of Traditional Statistics and VAEP

% ACCURATE

EXPAND

1	 J. Stones, 25  Manchester City	94.82
2	 A. Laporte, 25  Manchester City	94.2
3	 N. Otamendi, 32  Manchester City	94.19
4	 Eric García, 19  Manchester City	94.07
5	 Rodri Hernández, 23  Manchester City	93.64

Rank	Player	Team	Goals/90	Assists/90	VAEP/90
1	R. Mahrez	Manchester City FC	0.42	0.42	0.89
2	K. De Bruyne	Manchester City FC	0.31	0.59	0.79
3	S. Mané	Liverpool FC	0.57	0.24	0.69
4	S. Agüero	Manchester City FC	0.97	0.12	0.68
5	P. Aubameyang	Arsenal FC	0.62	0	0.62
6	D. Ings	Southampton FC	0.63	0.04	0.62
7	Mohamed Salah	Liverpool FC	0.61	0.08	0.61
8	J. Vardy	Leicester City FC	0.72	0.19	0.59
9	R. Sterling	Manchester City FC	0.46	0.04	0.59
10	Richarlison	Everton FC	0.36	0.07	0.56



Premier League 2019-20

Passing accuracy

79.7%



Have a conversation with earpiece!

https://earpiece.twelvefootball/conversation/46070

earpiece¹²

Menu

New chat

Today

Match Analyst

Barcelona vs Newcastle

First time using Earpiece? [Learn how it works](#) Give me the xG and xT map of Barca vs Newcastle match

To provide detailed information about the xG and xT map for the Barcelona vs. Newcastle match, I will redirect you to the Match Analyst. Please hold on.

In the match between Barcelona and Newcastle United, Barcelona had an xG of 1.62 and an xT of 1.30, while Newcastle United recorded an xG of 1.46 and an xT of 1.76. Notably, Barcelona had 9 shots with 2 on target, whereas Newcastle had 16 shots and 5 on target, showing a more efficient goal-scoring effort from Newcastle despite similar xG values.

31% 28% 41%
Draw

Goals 1 1
1.46 xG 1.62
48% Ball possession % 52%
16 Shots 9
5 Shots on target 2
9 Corners 4
6 Offsides 1
13 Fouls committed 9
2 Yellow cards 1
0 Red cards 0

Send a message to Earpiece

tarvir.anjum
Fred Plan

MATCH ANALYST

MATCH ANALYST